(a) Classical FJSP optimum, wear ignored: C_{max} = 29 (infeasible)
Machine 1 (T_1 = 10)
Machine 2 (T_2 = 14)
$O_{1,1}$ $O_{3,2}$ $O_{2,2}$ $O_{1,3}$ $O_{3,4}$ $O_{2,4}$ $O_{2,5}$
$O_{3,1}$ $O_{2,1}$ $O_{1,2}$ $O_{3,3}$ $O_{2,3}$ $O_{1,4}$
(b) Two-stage: sequences of (a), maintenance inserted just in time: C_{max} = 35
Machine 1 (T_1 = 10)
Machine 2 (T_2 = 14)
$O_{1,1}$ $O_{3,2}$ $O_{2,2}$ $O_{1,3}$ $O_{3,4}$ $O_{2,4}$ $O_{2,5}$
$O_{3,1}$ $O_{2,1}$ $O_{1,2}$ $O_{3,3}$ $O_{2,3}$ $O_{1,4}$
(c) Joint optimum (CP model): C_{max} = 33
Machine 1 (T_1 = 10)
Machine 2 (T_2 = 14)
$O_{1,1}$ $O_{3,2}$ $O_{2,2}$ $O_{3,4}$ $O_{2,4}$ $O_{1,3}$ $O_{2,5}$
$O_{3,1}$ $O_{2,1}$ $O_{3,3}$ $O_{2,3}$ $O_{1,2}$ $O_{1,4}$
0 5 10 15 20 25 30 35
Time
Job 1
Job 2
Job 3
Maintenance
Wear threshold first exceeded